

Integraltransformation Block 3

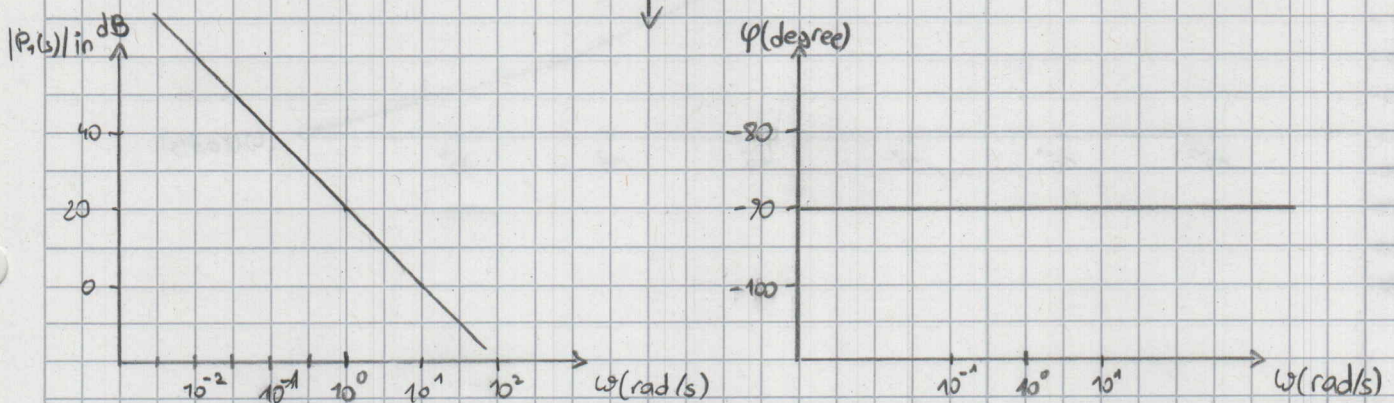
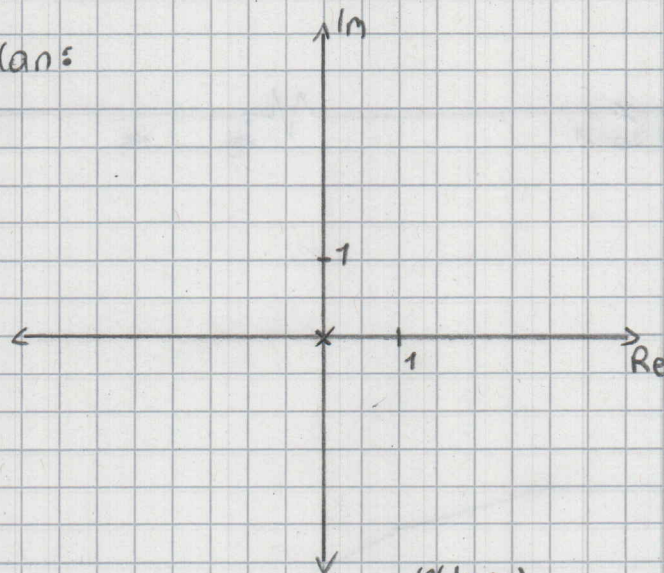
$P_1(s)$

geg.: $P_1(s) = \frac{10}{s}$

ges.: PN-Plan; Amplituden & Phasengang

Polstelle bei $x_1 = 0$

PN-Plan:



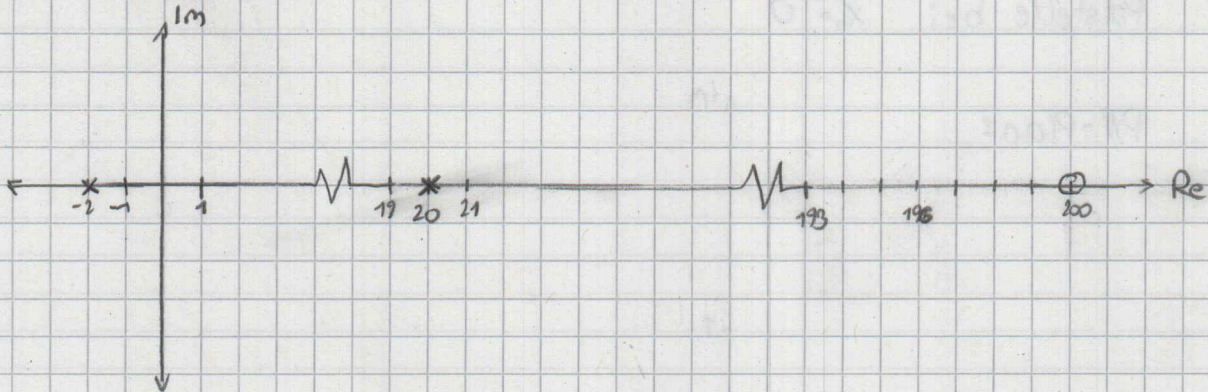
$P_2(s)$

geg.: $P_2(s) = \frac{1}{5} \frac{s-200}{(s+2) \cdot (s-20)}$

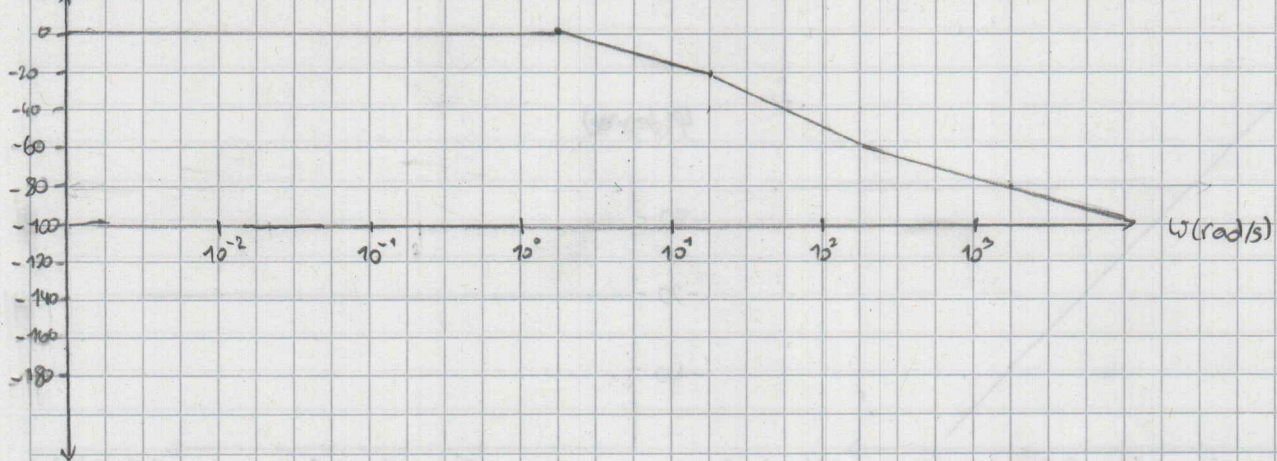
Pde: $x_1 = -2$
 $x_2 = 20$

Nullstelle: $n = +200$

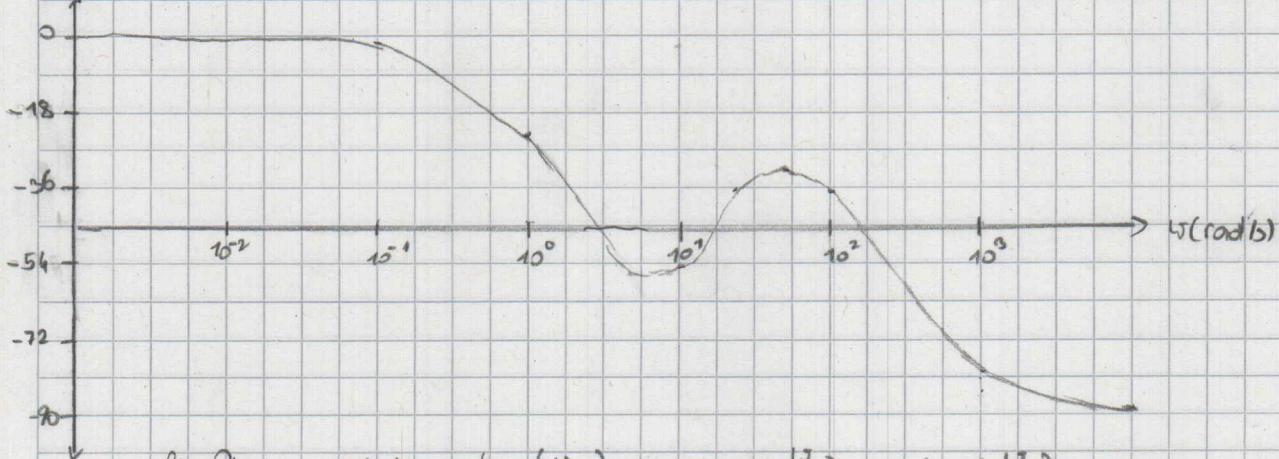
PN-Plan:



$|P_2(s)|$ in dB



$\varphi(\text{degree})$



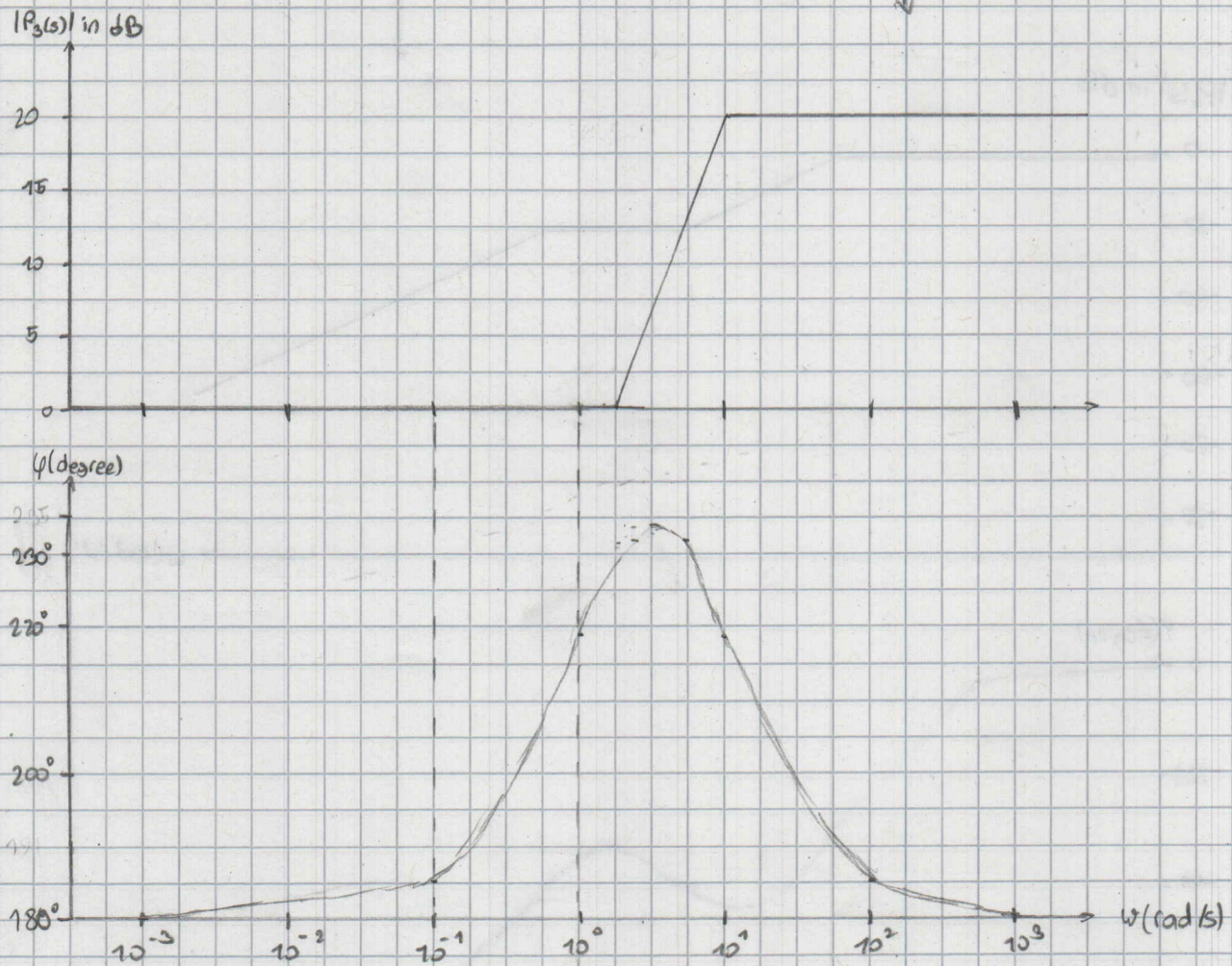
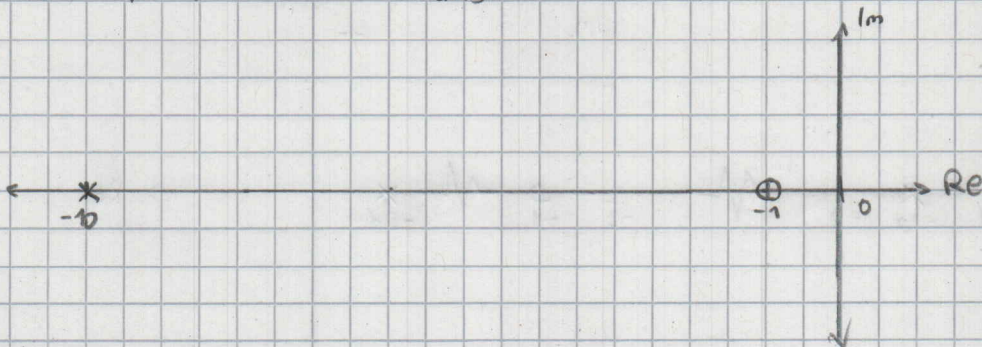
für Phase: $p(w) = \arctan\left(\frac{w}{-200}\right) - \arctan\left(\frac{w}{2}\right) - \arctan\left(\frac{w}{20}\right)$

$P_0(s)$

geg: $P_3(s) = \frac{-10 \cdot (s+1)}{s+10}$

Pole: $x_1 = -10$

Nullstelle: $n = -1$

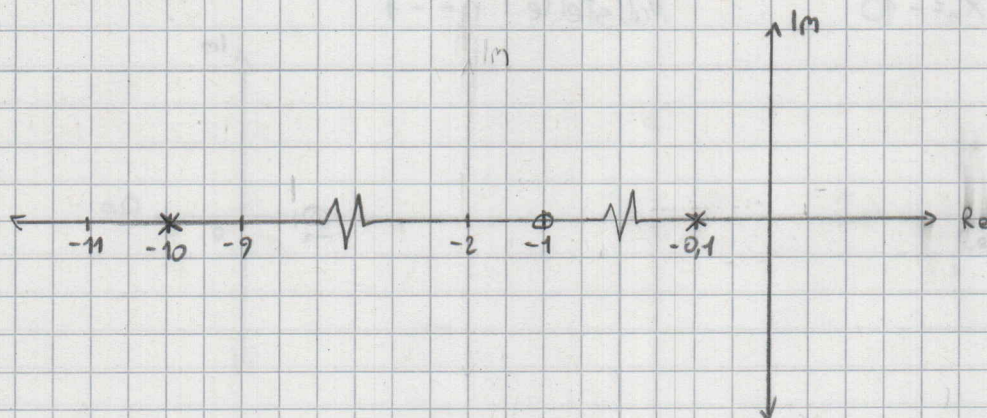


$P_4(s)$

geg.: $P_4(s) = \frac{s+1}{(s+0,1) \cdot (s+10)}$

Pole: $x_1 = -0,1$
 $x_2 = -10$

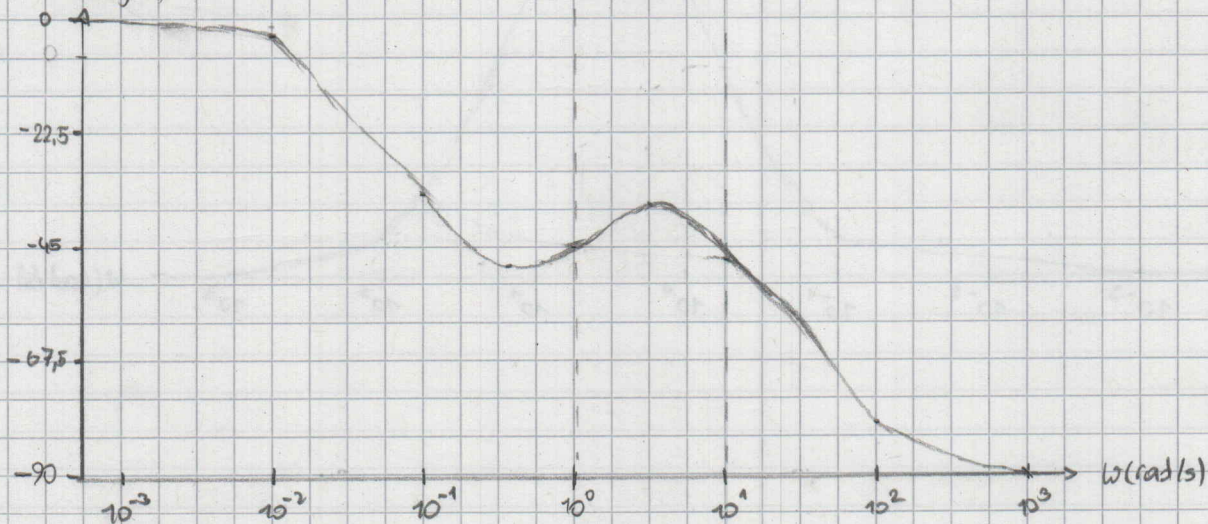
Nullstelle: $n = -1$



$|P_4(\omega)|$ in dB



φ (degree)

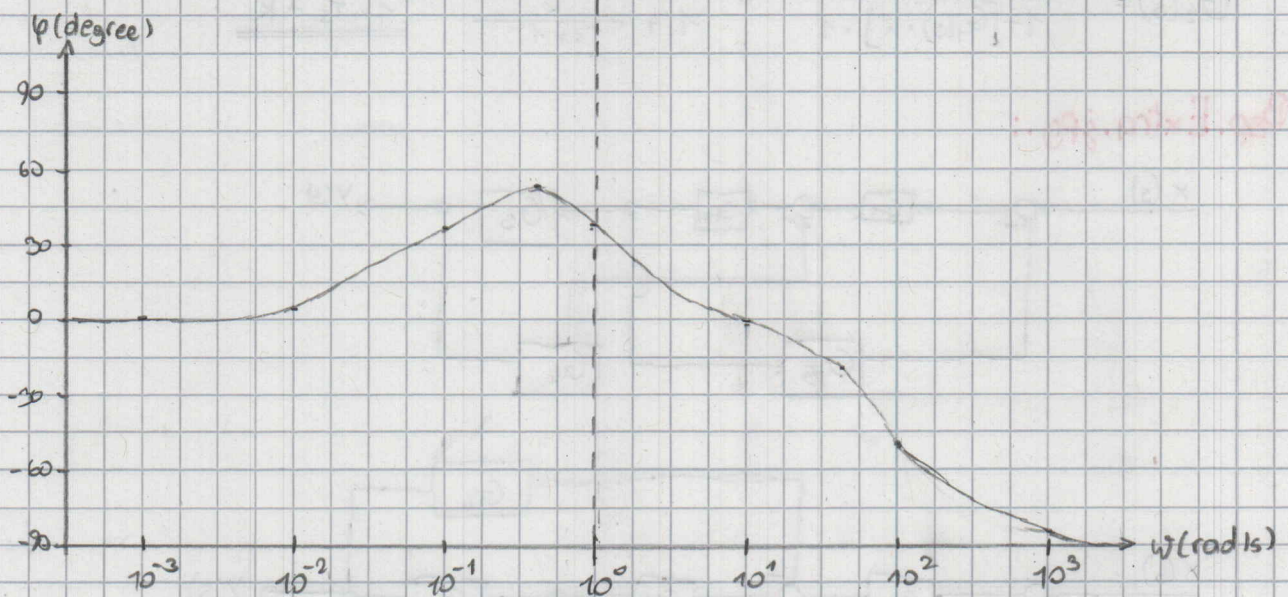
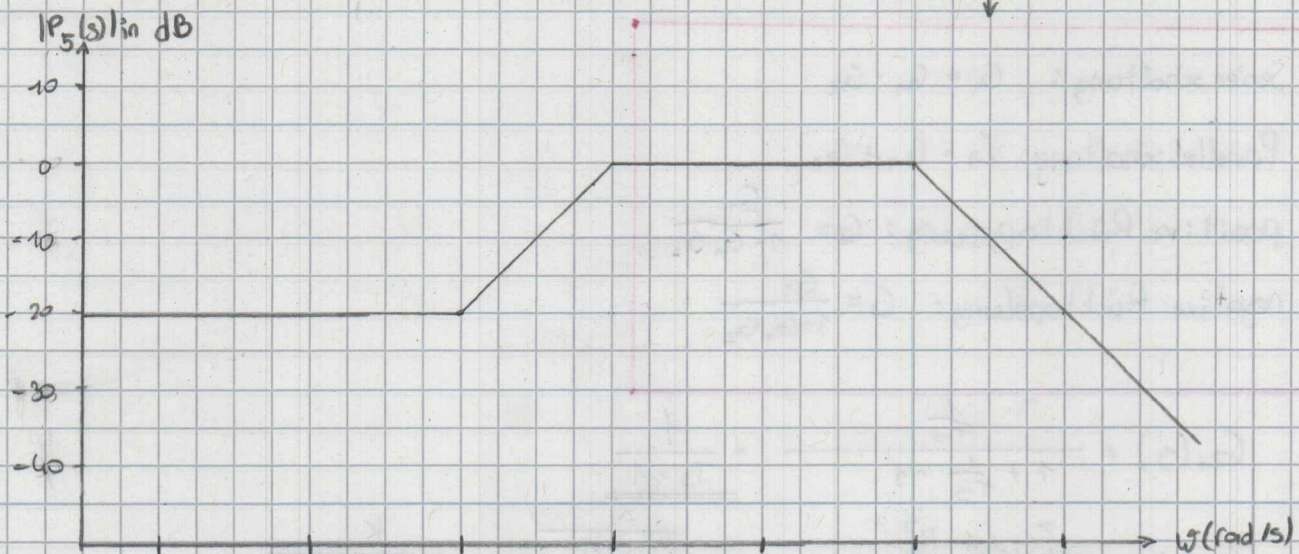
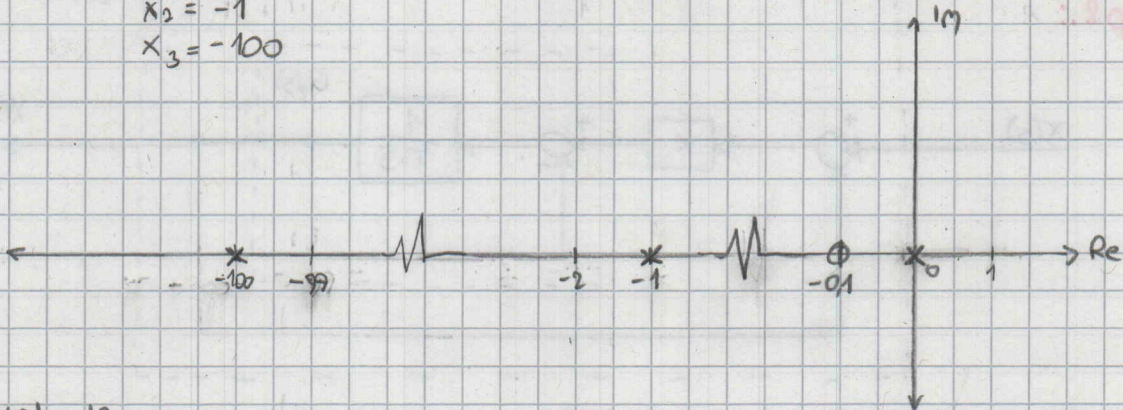


$P_5(s)$

geg.: $P_5(s) = \frac{100 \cdot (s+0,1)}{s \cdot (s+1) \cdot (s+100)}$

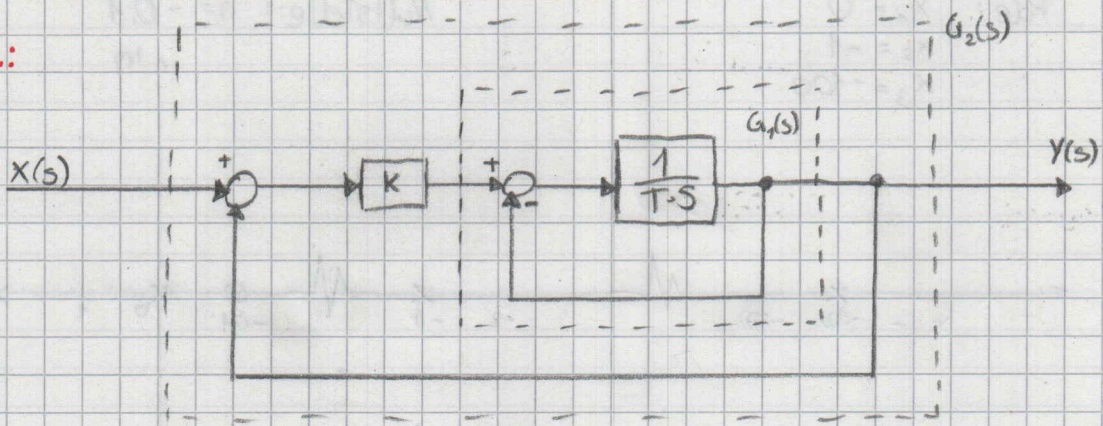
Pole: $x_1 = 0$
 $x_2 = -1$
 $x_3 = -100$

Nullstelle: $n = -0,1$



Blockschaltbild algebra:

Bsp 2.:



Serienschaltung: $G = G_1 \cdot G_2$

Parallelschaltung: $G = G_1 + G_2$

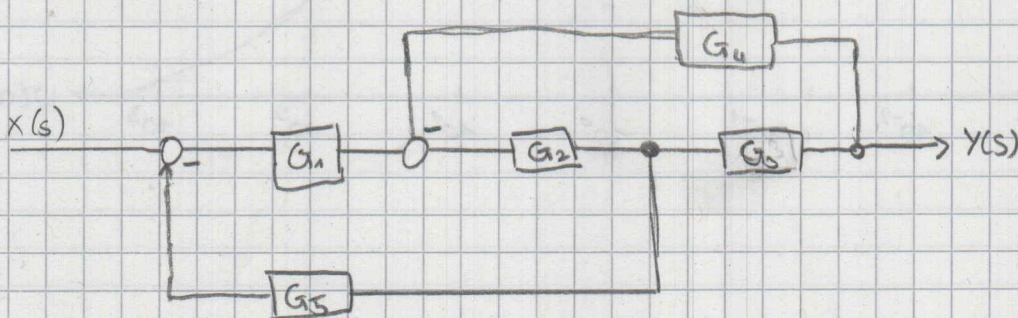
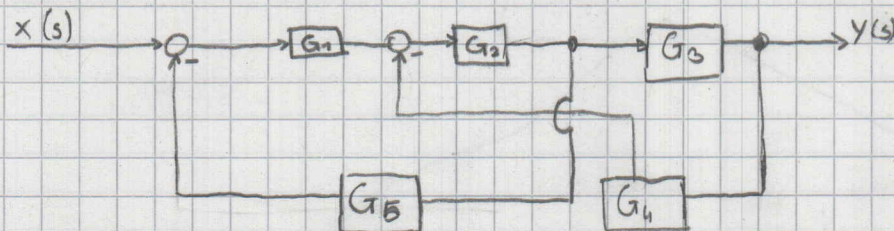
positive Rückkopplung: $G = \frac{G_1}{1 - G_1 G_2}$

negative Rückkopplung: $G = \frac{G_1}{1 + G_1 G_2}$

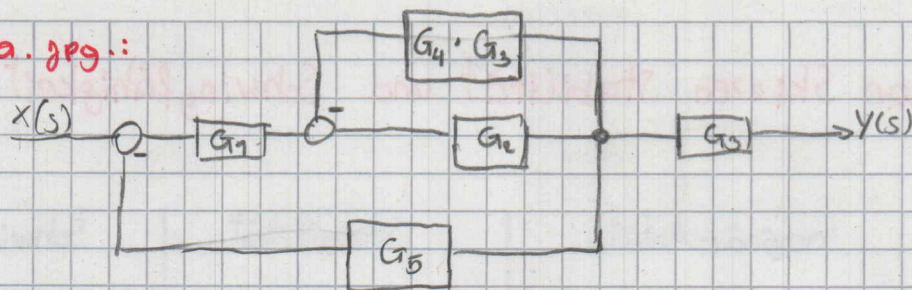
$$G_1(s) = \frac{\frac{1}{T \cdot s}}{1 + \frac{1}{T \cdot s} \cdot 1} = \frac{1}{T \cdot s + 1}$$

$$G_2(s) = \frac{[G_1(s) \cdot K]}{1 + [G_1(s) \cdot K] \cdot 1} = \frac{\frac{1}{T \cdot s + 1} \cdot K}{1 + \frac{K}{T \cdot s + 1}} = \frac{K}{1 + T \cdot s + K}$$

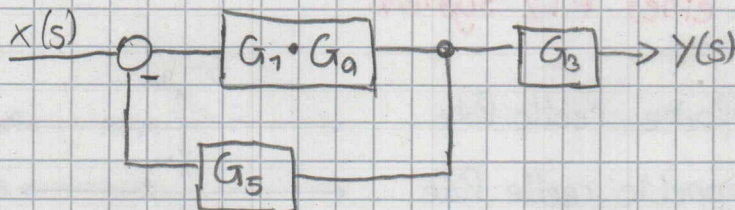
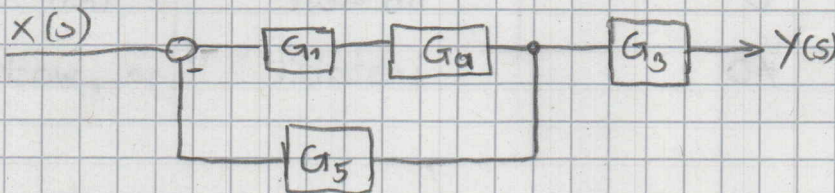
Bsp. Extra.jpg.:



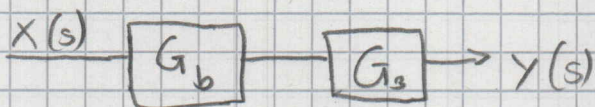
Extra. pg.:



$$G_a = \frac{G_2}{1 + G_2 \cdot [G_4 \cdot G_3]}$$



$$G_b = \frac{[G_1 \cdot G_a]}{1 + G_5 \cdot [G_1 \cdot G_a]}$$



$$G_{ges} = G_b \cdot G_3 = \frac{G_3}{G_2} \cdot \frac{G_1 \cdot G_a}{1 + G_5 \cdot G_1 \cdot G_a}$$

$$G_{ges} = G_3 \cdot \frac{G_1 \cdot \frac{1 + G_2 \cdot G_4 \cdot G_3}{G_2}}{1 + G_5 \cdot G_1 \cdot \frac{G_2}{1 + G_2 \cdot G_4 \cdot G_3}} = \frac{G_1 \cdot G_2 \cdot G_3}{1 + G_2 \cdot G_4 \cdot G_3 + G_5 \cdot G_1 \cdot G_2}$$

Mithilfe PN-Plan Skizzen Stabilität und Schwingfähigkeit beschreiben:

Real-Anteil	Imaginär-Anteil	Stabilität	Schwingfähigkeit
$Re < 0$	0	stabil	nein
$Re < 0$	$\neq 0$	stabil	ja, gedämpft
$Re = 0$	$\neq 0$	grenzstabil	ja, ungedämpft
$Re > 0$	0	instabil	nein
$Re > 0$	$\neq 0$	instabil	ja, wachsend

Mithilfe PN-Plan & Sprungantwort Skizzen Kriechfall / Aperiodischer Grenzfall / Schwingfall eines PT2 systems

Kriechfall:

Einfache reelle Pole

Aperiodischer Grenzfall: Doppelte reelle Pole

Schwingfall:

Konjugierte komplexe Polpaare

